

Hallucination Detection Report

Generated: 2026-04-16 10:36 · Extracted triples: 8 · Verdicts: 5 · Inconsistencies: 0 · Symbolic violations: 3

Executive Summary

2 Correct	1 Predicate Mismatch	2 Fabricated Fact
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Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
575	HAS_INTEREST_RATE	12.21%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the given subject.
580	HAS_INTEREST_RATE	11.91%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the given subject.
579	HAS_INTEREST_RATE	12.03%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the given subject.
579	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier classification for the given subject.
890	HAS_INTEREST_RATE	Below 4.36% (extrapolated, likely ~4.0% or lower)	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the given subject, including extrapolated or estimated values.
890	HAS_RISK_TIER	super prime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier classification for the given subject.
890	HAS_DOWN_PAYMENT	Below 2.0% (extrapolated, likely ~1.5% or lower)	Yes	ENTITY_ATTRIBUTE	Indicates the down payment requirement associated with the given subject, including extrapolated or estimated values.
335	HAS_LOAN_TERM	12 months	Yes	ENTITY_ATTRIBUTE	Indicates the loan term duration for the given subject.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	12.03%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_INTEREST_RATE	Below 4.36% (extrapolated, likely ~4.0% or lower)	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
890	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH

Subject	Original Pred.	Object	Canonical Pred.	Confidence
890	HAS_DOWN_PAYMENT	Below 2.0% (extrapolated, likely ~1.5% or lower)	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
335	HAS_LOAN_TERM	12 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	HIGH

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

Constraint ID	Rule Type	Severity	Subject	Predicate	Object	Explanation
c_interest_rate_range		MEDIUM	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	Below 4.36% (extrapolated, likely ~4.0% or lower)	The subject (credit score) is 890, which is greater than the allowed maximum of 850. According to the constraint, any subject > 850 is a violation.
c_tier_300_579		MEDIUM	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	The subject (credit score) is 890, which is outside the valid range 300-850. As per the constraint, a credit score above 850 always violates, regardless of the risk tier.
c_down_payment_range		MEDIUM	890	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Below 2.0% (extrapolated, likely ~1.5% or lower)	The subject (credit score) is 890, which is above 850. The constraint states that subject > 850 is always a violation, regardless of the down payment value.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT	STRICT		1.00	The subject and predicate match, and the object matches strictly with only a percentage symbol added. Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT	STRICT		1.00	The subject and predicate match, and the object matches strictly with only a percentage symbol added. Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	FABRICATED_FACT	NONE	SUBJECT	1.00	No DB evidence for subject and predicate; the triple is fabricated.

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
579	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	FABRICATED_FACT	NONE	SUBJECT	0.60 ■■■■■ ■■■	No DB candidate has subject 579 for this predicate and object, so the fact is fabricated. Evidence: subject: 375, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 380, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 385, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 390, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 395, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 400, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 405, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 410, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 415, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 420, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 425, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 430, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 435, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 440, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 445, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime
335	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	PREDICATE_MISMATCH	NONE	PREDICATE	0.60 ■■■■■ ■■■	Predicate not found for subject 335 in DB candidates, so predicate mismatch. Evidence: subject: 335, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 335, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 20.88

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
575	HAS_INTEREST_RATE	12.21%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT
580	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT
579	HAS_INTEREST_RATE	12.03%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	FABRICATED_FACT
579	HAS_RISK_TIER	deep subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	FABRICATED_FACT
335	HAS_LOAN_TERM	12 months	HIGH	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	PREDICATE_MISMATCH

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.